

Photosynthesis

Biology, Year 9, Unit 4: Plants

Learning objectives

- Describe what photosynthesis is and why plants need it
- Name the inputs and products of photosynthesis
- Explain the two stages: light reactions and the Calvin cycle
- Explain what limits the rate of photosynthesis

What is photosynthesis?

Plants are **producers**: they make their own food.

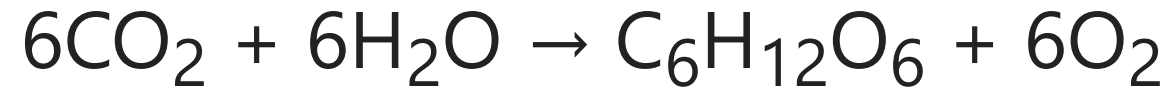
- They use energy from sunlight
- They turn water and carbon dioxide into glucose (a sugar)
- Oxygen is released as a by-product

The word equation

carbon dioxide + water → glucose + oxygen

Needs: light energy and chlorophyll

The symbol equation



Six molecules of carbon dioxide and six of water make one glucose and six oxygen.

What do plants take in?

- **Water:** absorbed by the roots, carried up in the xylem
- **Carbon dioxide:** enters the leaf through tiny holes called stomata
- **Light:** captured by the leaves

Inside a leaf

Upper
epidermis

Palisade layer (most
chloroplasts)

Spongy layer (air
spaces)

Lower epidermis with
stomata

Leaves are thin and flat to catch light and let gases move in and out.

Stomata and guard cells

- Stomata are tiny pores, mostly on the underside of the leaf
- Guard cells open and close them
- Open: carbon dioxide in, oxygen and water vapour out

Chloroplasts

- Photosynthesis happens inside chloroplasts
- Found in leaf cells, especially the palisade layer
- Contain stacks of membranes called thylakoids
- The fluid around them is called the stroma

Chlorophyll

- The green pigment inside chloroplasts
- Absorbs red and blue light best
- Reflects green light, so leaves look green



Blue: absorbed Green: reflected Red: absorbed

Two stages

1. Light reactions

Need light. Happen in the thylakoids.

2. Calvin cycle

Do not need light directly.
Happen in the stroma.

Stage 1: light-dependent reactions

- Light energy splits water molecules
- Oxygen is released (this is the oxygen we breathe)
- Energy is stored in carrier molecules ATP and NADPH

Stage 2: the Calvin cycle

- Also called the light-independent reactions
- Uses ATP and NADPH from stage 1
- Carbon dioxide is fixed into glucose

What do plants do with glucose?

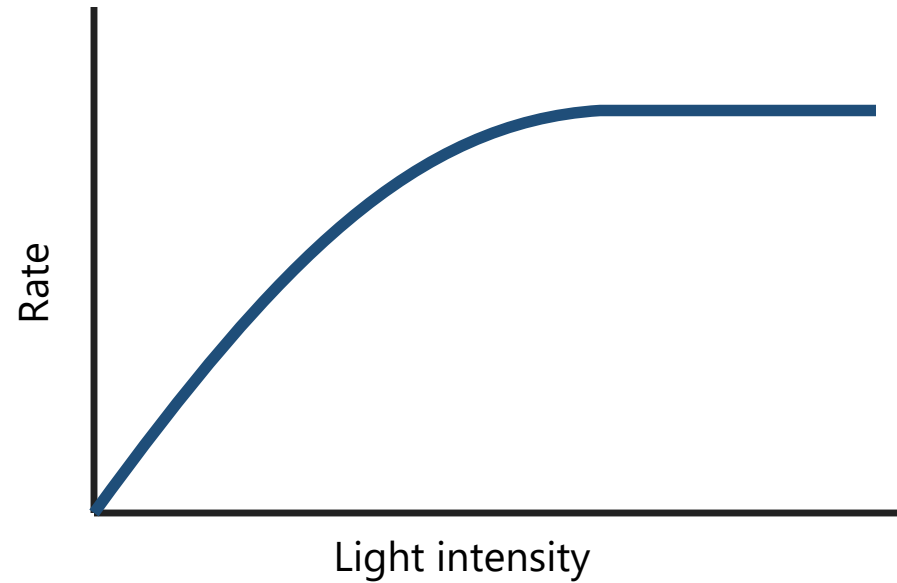
- Respiration: release energy for the plant
- Make cellulose to build cell walls
- Store it as starch for later

Why does it matter?

- Almost every food chain starts with a plant
- Puts oxygen into the air
- Removes carbon dioxide from the air

Limiting factors

- Light intensity
- Carbon dioxide concentration
- Temperature (enzymes)



When one factor goes up, the rate rises until another factor runs short.

Summary

- Plants use light to make glucose from carbon dioxide and water
- Chlorophyll in chloroplasts captures the light
- Light reactions release oxygen; the Calvin cycle builds glucose
- Light, carbon dioxide and temperature control the rate

Quick check

- Where does the oxygen released by plants come from?
- Why do leaves look green?
- Name one limiting factor.